Caregiving Robot: Tech Competition Brief & Build Overview

Target: NBQSA & Cutting Edge • 6-Month Timeline • Scope & Flexibility • 4-Tier Stack • Budget Breakdown • Key Differentiators

1. End Goal & Project Scope

- **Primary Objective & Competition Goal:** Compete and win in major national/international tech competitions and hackathons, specifically targeting **NBQSA** and **Cutting Edge**.
- **Project Scope:** Build an autonomous home-care robot and companion caregiver web app for elderly patients living alone.
- **Core Capabilities:** Automated pill & water dispensing, fall/unconsciousness detection, vitals monitoring, indoor navigation, and WebRTC family calls.
- **Development Timeline:** Strict **6-month delivery plan** to produce a fully functional, field-tested Phase 1 autonomous caregiving robot and companion app.
- **4-Tier Hybrid Architecture:**
    - **Tier 1 (ESP32-S3):** Real-time actuation (motors, stepper, pump) and low-level sensor input.
    - **Tier 2 (Raspberry Pi 4):** Onboard wireless bridge for webcam, speakerphone, LiDAR, and Smartwatch BLE.
    - **Tier 3 (Laptop Brain):** Heavy AI processing (YOLOv8-Pose, MediaPipe, Whisper STT, ROS 2 SLAM, FastAPI).
    - **Tier 4 (Caregiver Web App):** React-based dashboard for scheduling, live video, and Twilio emergency alerts.
- **Technology Stack Note:** While our core architectural stack is defined, specific hardware components and software tools remain **flexible and subject to refinement** as prototyping and testing progress.

2. What Makes Us Different (Competitive Advantage)

- **Commercial Off-The-Shelf Integration:** Eliminates costly custom health wearables by connecting directly to standard consumer smartwatches (Apple Watch / Wear OS) over BLE/Wi-Fi.
- **Fail-Safe 3-Step Pill Verification:** Combines physical proximity (PIR), physical passage (IR break-beam), and visual confirmation (MediaPipe Vision AI checking hand-to-mouth intake for 1.5s).
- **Open & Scalable Architecture:** Modular 4-tier stack allows offloading heavy computation to existing laptops or cloud instances, avoiding costly onboard industrial computing hardware.
    - **Local Cost Efficiency:** Fully functional MVP achieved under ~250,000 LKR total budget (including custom 3D printing & local BOM), delivering high competition value at a fraction of commercial costs.

2. Smartwatch Strategy & Indoor XYZ Positioning

Running on Commercial Smartwatches (Apple Watch vs. Wear OS)

- **No Custom Hardware Needed:** Companion software can run directly on off-the-shelf smartwatches to get heart rate and Z-axis fall spikes.
- **Apple Watch (watchOS):**
    - Use **CoreMotion API** to read 3-axis accelerometer and gyroscope data (detects sudden Z-axis downward drop).
    - Use **HealthKit / WorkoutSession** (or the free *HeartCast* app) to broadcast live Heart Rate (BPM) over standard Bluetooth BLE (UUID 0x180D) to the Raspberry Pi.
- **Wear OS (Samsung Galaxy Watch / Pixel Watch):**
    - Allows a custom Kotlin or Flutter background service to stream raw XYZ accelerometer packets and BPM over Wi-Fi or BLE directly to the laptop FastAPI server.

Solving the Indoor XYZ Position Challenge

- **Sensor Limitation:** Smartwatch accelerometers measure acceleration (G-force), not room-level coordinates; double-integrating acceleration for X/Y position drifts within

seconds.

- **Hybrid MVP Solution:**
  - **Z-Axis (Vertical Fall Spike):** Handled by the watch accelerometer (instant G-force spike when falling) + verified by the robot camera's YOLOv8-Pose bounding box aspect ratio flip.
  - **Rough Room Zone (Summon Mode):** Raspberry Pi measures **Bluetooth BLE RSSI signal strength** from the watch, or the patient uses voice/watch tap to summon the robot.
  - **Exact X/Y Position (Follow Mode):** Once in the room, the robot uses **2D 360° LiDAR + YOLOv8-Pose visual tracking** to lock onto the patient's exact X/Y distance and follow safely.

3. Core MVP Priorities (P1 – P6)

P1: Medicine Dispenser & Intake Confirmation

- **5-Cylinder Storage:** Holds 5 separate cylinders (each for a distinct medicine) plus an internal secure bottle/reservoir inaccessible to the patient.
- **Hand Placement Trigger:** A **PIR Sensor** mounted under the dispenser chute detects when the patient places their hand/arm near the cylinder outlet.
- **Exact Pill Release:** Stepper motor (NEMA 17 with A4988 or 28BYJ-48 with ULN2003) rotates the selected cylinder to drop the exact number of pills required for that dose.
- **Chute Drop Check:** An **IR Break-Beam Sensor** across the chute detects the exact millisecond each pill falls into the cup.
- **Visual Intake Confirmation:** Laptop runs **Google MediaPipe Hands + Pose**; when the patient's fingertips remain near their mouth landmarks for 1.5 seconds after pill drop, the dose is marked with a tick (✓) in the monitoring log.

P2: Voice & Interaction

- **Audio Hardware:** USB conference speakerphone on the robot provides a far-field microphone array and speaker with hardware echo cancellation.
- **Speech-to-Text (STT):** Patient speech is streamed to the laptop and transcribed in ~0.3 seconds using **Faster-Whisper**.

- **Conversational Brain:** LLM processes natural voice commands (e.g., *"Robot, come here"*, *"Call my daughter"*, *"What time is my medicine?"*).
- **Text-to-Speech (TTS): Piper TTS** speaks reminders, companionship responses, and walking guidance cues.

P3: WebRTC Audio/Video Calls

- **Video Pipeline:** 1080p USB webcam on the Raspberry Pi streams compressed 720p @ 20fps H.264 video via **MediaMTX** and **LiveKit WebRTC**.
- **Zero-Lag Two-Way Calling:** Caregivers tap "Call Patient" on the web app to talk and see through the robot camera; the patient talks hands-free.
- **Permission System:** Primary caregiver can call anytime; secondary caregivers (relatives, cousins, friends) can call only when granted permission by the primary caregiver.

P4: Autonomous Mobility & Navigation

- **High-Torque Drive:** Two 12V DC geared motors with wheel encoders driven by a **BTS7960 43A H-Bridge Motor Driver**.
- **Laser Mapping:** 2D 360° LiDAR (LD19 or RPLIDAR A1M8) scans walls and furniture, feeding **ROS 2 SLAM Toolbox & Nav2** on the laptop.
- **Operation Modes:**
    - **Follow Mode:** Tracks and travels with the patient at a safe 1.2m–1.5m distance.
    - **Summon ("Call") Mode:** Drives across the floor from another room when called by the patient.
    - **Visual Guidance Mode:** Drives at a gentle 0.4 m/s pace while giving spoken walking cues.
- **500ms Safety Watchdog:** ESP32 hardware timer automatically brakes both wheels to 0 PWM if Wi-Fi communication drops for more than 0.5 seconds.

P5: Health Vitals & Emergency Fall Detection

- **Live Heart Rate:** Smartwatch broadcasts continuous BPM over Bluetooth BLE to the Raspberry Pi.
- **Contactless Body Temperature: MLX90614 IR Thermometer** next to the dispenser reads wrist/forehead temperature in 0.5 seconds during pill pickup.
- **Dual-Confirmed Fall Detection:** Combines Smartwatch Z-axis drop spike with **YOLOv8n-Pose** (detects person bounding box flipping from vertical to horizontal near the

floor).

- **Unconsciousness Distress Call:** If motionless for 10 seconds, the robot announces: *"Fall detected. Say cancel if you are okay."* If no response within 5 seconds, **Twilio API** automatically calls and texts the caregiver's +94 mobile number with an emergency live camera link.

P6: Automated Water Dispensing

- **Reservoir:** Food-grade sealed water tank mounted low in the robot chassis for stability.
- **Pump Actuation:** 5V/12V peristaltic or submersible mini water pump controlled via an ESP32 relay/MOSFET module (GPIO 15).
- **Synchronized Delivery:** Automatically dispenses a pre-set volume of water into a cup alongside scheduled pill dispensing or upon voice request.

4. 4-Tier Hardware & Pinout Map

Tier	Device	Role	Connected Modules & Exact Pinout
Tier 1	**ESP32-S3** *(Onboard)*	Real-time motor control, pill dispensing, water pump, and low-level sensors	• **BTS7960 Motor Driver:** GPIO 25, 26, 27, 14 (PWM) • **Wheel Encoders:** GPIO 32, 33, 34, 35 • **PIR Hand Sensor:** GPIO 5 (Digital In) • **5-Cylinder Pill Stepper:** GPIO 18 (STEP), GPIO 19 (DIR) • **IR Pill-Drop Sensor:** GPIO 4 (Digital In)

Tier	Device	Role	Connected Modules & Exact Pinout
			• **Water Pump Relay:** GPIO 15 (Digital Out) • **MLX90614 IR Temp Sensor:** I2C GPIO 21 (SDA), GPIO 22 (SCL)
Tier 2	**Raspberry Pi 4 (4GB)** *(Onboard)*	High-bandwidth USB sensor hub and 5GHz Wi-Fi bridge	• **ESP32 Serial Link:** USB Serial (/dev/ttyUSB0 @ 115200 baud) • **1080p Webcam:** USB 3.0 Port • **Conference Speakerphone:** USB 2.0 Port • **2D 360° LiDAR (LD19/A1):** USB-UART • **Smartwatch Link:** Onboard Bluetooth BLE (UUID 0x180D)
Tier 3	**Laptop / PC** *(Offboard via Wi-Fi)*	Heavy GPU AI inference, ROS 2 SLAM navigation, and FastAPI backend	• **Fall & Pose Tracking:** YOLOv8n-Pose (Python / GPU) • **Pill Intake Verification:**

Tier	Device	Role	Connected Modules & Exact Pinout
			MediaPipe Hands + Pose • **Navigation Stack:** ROS 2 Humble/Jazzy (SLAM Toolbox + Nav2) • **Voice Pipeline:** Faster-Whisper STT + LLM + Piper TTS • **Backend Server:** Python FastAPI + WebSockets + Twilio API
Tier 4	**Caregiver Web App** *(Mobile / Desktop)*	Remote family dashboard, scheduling, and live calling	• **Frontend:** React Web App (via Cloudflare/Ngrok tunnel) • **Live Video/Audio:** LiveKit WebRTC • **Emergency Alerts:** WebSockets + Push + Twilio (+94 Call/SMS)

5. Companion Software & Main API Endpoints

Core Software Rules

- **Medicine Schedule Entry:** Primary caregiver enters Medicine Name, Cylinder ID (1–5), daily frequency, exact dose times, and pills per dose.
- **Pill-Level Inventory Tracking:** Software automatically deducts dispensed pills from cylinder stock and alerts the primary caregiver when a cylinder is running low.
- **Missed-Dose Escalation Chain:** If a scheduled dose is not confirmed with a tick (✓) within the grace window:
    1. **Alarm:** Robot sounds an onboard voice and chime alarm.
    2. **Notification:** App pushes an immediate alert to the caregiver's phone.
    3. **Call Function:** Prompts caregiver to initiate a live WebRTC video call.
- **Caregiver Roles:** Primary caregiver manages the system and can grant permission to a secondary/external caregiver (friend, cousin, relative) to receive alerts and video-call the patient.

FastAPI Main Endpoints

#	Endpoint & Method	Feature	Payload / Parameters	System Action
1	POST /api/notifications/push	Notification Push	type (fall / missed_dose / low_pills), severity, patient_id	Sends push alert to app; if type == "fall", triggers Twilio SMS + auto-dial to +94 phone.
2	POST /api/call/live-session	Call Function & Live Camera Feed	caregiver_id, role_token, audio_enabled	Verifies primary or permitted secondary caregiver role and returns LiveKit WebRTC session token.

#	Endpoint & Method	Feature	Payload / Parameters	System Action
3	GET /api/schedule/status POST /api/schedule/escalate	Schedule: Completed / Not Completed	date, dose_id, status (completed / pending / missed)	Returns tick (✓) log. If not completed, triggers: **1) Robot Alarm → 2) Push Notification → 3) Call prompt**.
4	POST /api/medicines/configure	Medicine Name & Dispense Rules	cylinder_id (1-5), medicine_name, times_per_day, dose_times[], pills_per_dispense, initial_stock	Saves schedule, syncs timers with ESP32 carousel, and initializes pill-level inventory tracking.
5	POST /api/caregivers/permissions	External / Secondary Caregiver Access	primary_id, secondary_name, phone (+94), can_call (bool)	Allows primary caregiver to suggest/authorize a secondary caregiver for alerts and video calling.
6	WS /ws/robot/telemetry POST	Live Vitals, Water & Robot Summon	heart_rate_bpm, wrist_temp_c, xyz_status, target_room,	Streams live Apple Watch BPM, MLX90614 temp, cylinder

#	Endpoint & Method	Feature	Payload / Parameters	System Action
	/api/robot/summon		dispense_water (bool)	stock, and triggers room summon or water pump.

6. System Deliverables & Implementation Phases

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  Phase 1: 5-Cylinder Pill Dispenser, PIR Trigger, Water Pump & Temp Box
  - Wire PIR hand sensor (GPIO 5), stepper motor (GPIO 18, 19), IR break-beam sensor (GPIO 4), water pump relay (GPIO 15), and MLX90614 thermometer (GPIO 21, 22) to the ESP32-S3.
  - Assemble the 5-cylinder pill carousel and test exact multi-pill drops and water dispensing on the bench.
- **Phase 2: Vision AI Pipeline on Laptop**
  - Set up Google MediaPipe Hands + Pose on the laptop to verify hand-to-mouth pill intake (1.5s threshold).
  - Set up YOLOv8n-Pose to detect vertical-to-horizontal falls and 10-second motionlessness.
  - Connect webcam and speakerphone to the Raspberry Pi 4 and stream compressed 720p H.264 video over Wi-Fi via MediaMTX.
- **Phase 3: Rolling Chassis, BTS7960 Driver & Safety Watchdog**
  - Wire two 12V DC geared encoder motors through the **BTS7960 43A motor driver** and power via 12V battery + LM2596 buck converters.
  - Connect ESP32 to Raspberry Pi via USB serial (/dev/ttyUSB0) and program the 500ms Wi-Fi safety watchdog brake.
- **Phase 4: 2D LiDAR, Autonomous Navigation, Summon & Follow Modes**
  - Mount LD19 or RPLIDAR A1M8 on the robot and stream /scan data to ROS 2 SLAM Toolbox & Nav2 on the laptop.

    - Map the house floorplan, label room zones for Summon Mode, and test YOLOv8 + LiDAR Patient Follow Mode.
- **Phase 5: Smartwatch BLE, Voice AI & Caregiver Web App**
    - Connect Smartwatch BLE Heart Rate (UUID 0x180D) and Z-axis fall spikes to the Raspberry Pi.
    - Integrate Faster-Whisper STT + LLM + Piper TTS for hands-free voice interaction.
    - Deploy the FastAPI + React caregiver app with LiveKit WebRTC video calls, missed-dose escalations, and Twilio +94 emergency dialing.

7. Sri Lanka Bill of Materials & Cost Breakdown (LKR)

Option A: MVP Using Existing Laptop as the AI Brain (Recommended)

Component / Part	MVP Purpose	Sri Lanka Suppliers	Estimated Price (LKR)
Raspberry Pi 4 Model B (4GB) + SD + PSU	Onboard wireless bridge for Webcam, Speakerphone, BLE & LiDAR	Tronic.lk / Circuit.lk / Scion	Rs. 38,000 – 58,000
ESP32-S3 Dev Board	Real-time control for BTS7960, encoders, stepper, PIR, IR & water pump	Tronic.lk / Microchip.lk	Rs. 1,630 – 2,400
2D 360° LiDAR (LD19 or RPLIDAR A1M8)	360° laser room mapping, obstacle avoidance & summon	Daraz.lk / AliExpress SL Direct	Rs. 26,000 – 32,000

Component / Part	MVP Purpose	Sri Lanka Suppliers	Estimated Price (LKR)
	navigation (P4)		
2x 12V DC Geared Motors + Encoders + Wheels	Drives the robot chassis smoothly across rooms	Tronic.lk / Arrow Electronics SL	Rs. 9,500 – 12,500
BTS7960 43A Motor Driver + LM2596 Buck	Upgraded high-current motor driver + 12V-to-5V power regulation	Tronic.lk / Scion Electronics	Rs. 2,200 – 3,200
Stepper Motor + Driver + PIR + IR Break-Beam	5-cylinder pill dispenser actuation, hand detection & pill drop check (P1)	Tronic.lk / Circuit.lk	Rs. 1,800 – 4,000
Mini 5V/12V Water Pump + Relay + Tubing	Automated water dispensing system for the patient (P6)	Tronic.lk / Daraz.lk	Rs. 1,200 – 2,200
MLX90614 Contactless IR Temp Sensor	Checks patient wrist/forehead temperature during pill pickup (P5)	Circuit.lk / Daraz.lk	Rs. 2,600 – 3,800

Component / Part	MVP Purpose	Sri Lanka Suppliers	Estimated Price (LKR)
1080p USB Webcam + USB Speakerphone	Streams video for YOLO/MediaPipe, voice AI (P2) & WebRTC calls (P3)	Barclays / Nanotek / Daraz.lk	Rs. 12,000 – 16,500
12V 7Ah Battery (or 3S Li-ion) + Charger + Frame	Powers robot for 3–5 hours + 5-cylinder acrylic/3D-printed body	Local Battery Shop / 3D Print SL	Rs. 11,500 – 15,000
Existing Laptop / PC + Existing Smartwatch	Runs YOLOv8, MediaPipe, Whisper, ROS 2 SLAM, FastAPI & BLE Vitals	Already Owned	Rs. 0
3D Printing & Custom Prototyping Allocation			**Rs. 100,000**

TOTAL ESTIMATED MVP COST (WITH 3D PRINTING & LAPTOP BRAIN)	**Rs. 206,430 – Rs. 249,600**

Option B: Standalone Robot Upgrade (No Laptop Needed)

- **Replace Raspberry Pi 4 + Laptop with:** Raspberry Pi 5 (8GB) + Raspberry Pi AI HAT+ (Hailo-8L 13 TOPS NPU) or NVIDIA Jetson Orin Nano (Rs. 127,000 – 230,000) and a higher-capacity 3S/4S LiFePO4 battery pack (Rs. 18,000 – 25,000).
- **Total Standalone Estimated Cost: Rs. 201,930 – Rs. 331,600**.